

TRACELESS $SU(2)$ CHARACTERS AND $\mathbb{Z}/4$ INSTANTON GRADINGS FOR TWO-BRIDGE AND $(3, n)$ -TORUS KNOTS

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ABSTRACT. We assemble, and where possible independently verify, the representation-theoretic data underlying the pillowcase (symplectic) side of the Atiyah–Floer conjecture for knots, for two families: the two-bridge knots and the $(3, n)$ -torus knots. For a two-bridge knot $\mathfrak{b}(p, q)$ we give a short, self-contained proof that every irreducible traceless $SU(2)$ representation is binary-dihedral; these are the $(p-1)/2$ dihedral characters, at meridian-pair angles ω with $\cos \omega = \cos(2\pi k/p)$, independent of q , and the traceless Riley polynomial is the explicit product $\phi_p(u) = \prod_k (u + 4 \sin^2(\pi k/p))$, monic of degree $(p-1)/2$ with constant term $\det K$. We use this to give a transparent account of the Hedden–Herald–Kirk theorem that pillowcase homology equals reduced singular instanton knot homology $I^\natural$ on this family, and of why the figure-eight bubbling and bounding cochains that obstruct the general conjecture are structurally inert there. For the $(3, n)$ -torus knots we compute the full traceless character variety and prove a dihedral dichotomy: exactly $(\det -1)/2$ characters are dihedral, where $\det$ is the knot determinant, so for n odd every irreducible traceless character is *non-dihedral*. Passing to the double branched cover $\Sigma(2, 3, n)$, we compute the $\mathbb{Z}/4$ spectral-flow gradings of the generators directly from the Fintushel–Stern index and the equivariant ρ -invariant, independently of, and checked against, the Poudel–Saveliev and Anvari computations; for n odd the gradings split evenly between 1 and 3, giving the singular-instanton *chain complex* $IC^\natural(T(3, n)) = (1 + a, a, a, a)$ (ranks in gradings $0, 1, 2, 3$), $a = -\sigma/4$. The *homology* $I^\natural$ equals this for $n \equiv 1 \pmod{6}$ (differential zero) but is smaller by 2 for $n \equiv 5$, where the differential is nonzero: $\text{rank } I^\natural = \sum_i |\Delta_{T(3, n)}|$, the sum of the absolute values of the Alexander coefficients, throughout, and $T(3, 5) = P(-2, 3, 5) = 10_{124}$ has rank 7, not the chain rank 9. Finally we reproduce the first nonzero pillowcase differential, for $8_{19} = T(3, 4)$, and identify it as the corner figure-eight bigon that is absent on two-bridge knots. Throughout we distinguish carefully between results proved here, classical facts, and computations reproduced from the literature.

1. INTRODUCTION

1.1. The Atiyah–Floer conjecture and its knot version. Let Y be a closed oriented 3-manifold with a Heegaard splitting $Y = Y_0 \cup_\Sigma Y_1$. Two Floer theories attach to this picture. On the gauge-theoretic side is Floer’s instanton homology $HF^{\text{inst}}(Y)$, generated by flat $SU(2)$ connections and with a differential counting anti-self-dual instantons on $Y \times \mathbb{R}$. On the symplectic side, the moduli space $M(\Sigma)$ of flat $SU(2)$ connections on Σ is symplectic, the handlebodies determine Lagrangians $L_0, L_1 \subset M(\Sigma)$, and one forms the Lagrangian Floer homology $HF^{\text{symp}}(L_0, L_1)$. The Atiyah–Floer conjecture [Ati88] asserts these are isomorphic (for a recent survey of the Lagrangian Floer theory in which the conjecture sits, see Auroux [Aur25]). The one general theorem is the mapping-torus case of Dostoglou–Salamon [DS94]; the Heegaard-splitting statement remains open, obstructed by the singularities of $M(\Sigma)$ and by figure-eight bubbling in the adiabatic limit.

There is a knot-theoretic analogue in which the pillowcase plays the role of $M(\Sigma)$. Let $K \subset S^3$ be a knot, and cut S^3 along a Conway sphere meeting K in four points into two 2-string tangles.

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A representation $\rho: \pi_1(S^3 \setminus K) \rightarrow \mathrm{SU}(2)$ is *traceless* if every meridian maps to a traceless element (conjugate to $\mathrm{diag}(i, -i)$); this is the boundary condition of Kronheimer–Mrowka’s singular instanton theory [KM11a, KM11b]. Restricting traceless flat connections of each tangle to the four-punctured sphere and applying holonomy perturbations, one obtains immersed Lagrangians L_0, L_1 in the *pillowcase*

$$P = (\mathbb{R}/2\pi\mathbb{Z})^2 / \iota, \quad \iota(\gamma, \theta) = (-\gamma, -\theta),$$

the traceless $\mathrm{SU}(2)$ character variety of the four-punctured sphere: a two-sphere with four $\mathbb{Z}/2$ orbifold points [Lin92, HHK14]. The *knot Atiyah–Floer conjecture* of Hedden–Herald–Kirk and Cazassus–Herald–Kirk–Kotelskiy [HHK14, CHKK22] asserts that a bounding-cochain-deformed Lagrangian Floer homology of (L_0, L_1) is isomorphic, as a relatively $\mathbb{Z}/4$ -graded group, to the reduced singular instanton knot homology $I^\natural(K)$. (A *bounding cochain* is a formal combination of self-intersection points of an immersed Lagrangian that deforms the Floer differential by higher-polygon counts — the standard device by which Lagrangian Floer theory absorbs disk bubbling [Aur25, CHKK22]; *figure-eight bubbling* is the degeneration of holomorphic strips, specific to immersed Lagrangians, that makes the correction necessary here [CHKK22].)

1.2. Purpose and scope. This paper is a computational and expository consolidation. Its aim is to make explicit, for two tractable infinite families, the representation-theoretic data on which the pillowcase construction rests: the traceless character variety (the generators), the $\mathbb{Z}/4$ spectral-flow gradings, and the differential. The same data is in active use beyond the pillowcase program: Dai–Mallick–Taniguchi have recently used an explicit analysis of the traceless character varieties of two-bridge knots to compute, in part, symmetry actions on their singular instanton complexes [DMT26]. We are careful throughout to separate three kinds of statement:

- (i) results proved here (self-contained, though some are surely classical);
- (ii) classical facts and standing theorems (attributed);
- (iii) computations reproduced from [HHK18, PS17, Anv16] (attributed, and independently checked where possible).

We do *not* claim any new theorem about the knot Atiyah–Floer conjecture itself: for two-bridge knots it is a theorem of Hedden–Herald–Kirk [HHK14, HHK18], and for the torus knots we treat the instanton side and the character-variety side, not the analytic identification of the two. The explicit verifications reported below—the exact-arithmetic check of Theorem 1.1, the counts of Proposition 4.1, and the gradings of Section 5—are carried out by short computer programs, included as ancillary files with this preprint and also available at the public repository <https://github.com/bwuebben/pillowcase-bounding-cochain>. Each is a short self-contained Python program that checks its own output; the first works in exact Gaussian-integer arithmetic.

1.3. Results. Section 2 records the pillowcase and the traceless characters as generators. Section 3 proves:

Theorem 1.1 (Dihedral rigidity for two-bridge knots). *Let $K = \mathfrak{b}(p, q)$ be a two-bridge knot, $p = \det K$. Every irreducible traceless $\mathrm{SU}(2)$ representation of $\pi_1(S^3 \setminus K)$ is binary-dihedral. There are exactly $(p-1)/2$ of them: writing ω for the meridian-pair angle (the angle between the axes of two chosen meridians, the conjugacy invariant of Section 2), they occur at $\cos \omega = \cos(2\pi k/p)$ for $k = 1, \dots, (p-1)/2$, independent of q ; equivalently the traceless Riley polynomial in $u = 2 \cos \omega - 2$ is*

$$\phi_p(u) = \prod_{k=1}^{(p-1)/2} \left(u + 4 \sin^2 \frac{\pi k}{p} \right),$$

monic of degree $(p-1)/2$, with integer coefficients, $\phi_p(0) = p = \det K$, and root-sum $-p$.

We then explain (Section 3.2) how Theorem 1.1 clarifies the Hedden–Herald–Kirk two-bridge isomorphism and localizes the obstruction, and we correct a naive expectation about the earring correspondence.

Section 4 computes the $(3, n)$ -torus knot characters and proves:

Theorem 1.2 (Dihedral dichotomy for $(3, n)$ -torus knots). *Let $\gcd(3, n) = 1$. The irreducible traceless $\mathrm{SU}(2)$ characters of $T(3, n)$ are in bijection with the admissible representation arcs of Section 4 (the arcs of irreducible characters that contain a traceless point), one per arc. Exactly $(\det - 1)/2$ of them are binary-dihedral: one if n is even ($\det = 3$) and none if n is odd ($\det = 1$). In particular, for n odd every irreducible traceless character of $T(3, n)$ is non-dihedral.*

Section 5 passes to the double branched cover $\Sigma(2, 3, n)$ and computes the $\mathbb{Z}/4$ gradings directly, obtaining for n odd the chain complex $IC^\natural(T(3, n)) = (1 + a, a, a, a)$, $a = -\sigma/4$, whose homology $I^\natural$ equals it for $n \equiv 1 \pmod{6}$ but is smaller by 2 for $n \equiv 5 \pmod{6}$ (where the differential is nonzero); the gradings split evenly between $\mu = 1$ and $\mu = 3$, which yields the chain-level ranks (Proposition 5.2). Section 6 reproduces the 8_{19} differential and identifies its geometric source. Section 7 states the genuinely open direction.

Acknowledgements. The literature computations reproduced here are due to Hedden–Herald–Kirk [HHK14, HHK18], Cazassus–Herald–Kirk–Kotelskiy [CHKK22], Poudel–Saveliev [PS17], and Anvari [Anv16]; we have tried to attribute each precisely.

2. THE PILLOWCASE AND TRACELESS CHARACTERS

We work with $\mathrm{SU}(2)$ as the unit quaternions. An element is traceless iff it is a purely imaginary unit quaternion (eigenvalues $\pm i$, squaring to -1); as a rotation it is a rotation by π . The traceless elements form a 2-sphere $S \subset \mathrm{Im} \mathbb{H} \cong \mathbb{R}^3$.

Proposition 2.1 (Classical; cf. [Lin92, HHK14]). *The traceless $\mathrm{SU}(2)$ character variety of the four-punctured sphere is the pillowcase $P = (\mathbb{R}/2\pi\mathbb{Z})^2/\iota$, with four $\mathbb{Z}/2$ orbifold points at $(\gamma, \theta) \in \{0, \pi\}^2$ and the symplectic form descending from $d\gamma \wedge d\theta$.*

Sketch. Write π_1 of the four-punctured sphere as $\langle X_1, \dots, X_4 \mid X_1 X_2 X_3 X_4 = 1 \rangle$ with each $X_i \in S$. Since purely imaginary units satisfy $X_i^{-1} = -X_i$, the relation gives $X_1 X_2 = X_4 X_3 =: A$. For pure imaginary units, $X_i X_j = -\langle X_i, X_j \rangle + X_i \times X_j$, so when $A \neq \pm 1$ its axis is well defined; equating the two expressions for A forces the four X_i to be coplanar, orthogonal to axis(A). Placing that axis along k and writing $X_m = \cos \beta_m i + \sin \beta_m j$, the relation becomes $\beta_2 - \beta_1 = \beta_3 - \beta_4$; gauge-fixing $\beta_1 = 0$ and setting $\gamma = \beta_2$, $\theta = \beta_4$ leaves two parameters, and the residual conjugation by the π -rotation about i acts by $(\gamma, \theta) \mapsto (-\gamma, -\theta)$. The four fixed points are the reducible (collinear) characters. $\square$

For a knot K with a two-tangle decomposition, a representation of $\pi_1(S^3 \setminus K)$ is a compatible pair of tangle representations, i.e. a point of $L_0 \cap L_1$. Thus the intersection points are exactly the traceless representations of the knot group, and (after perturbation) the generators of the singular instanton chain complex [KM11a, KM11b]. In what follows a representation is determined up to conjugacy by the single invariant $\cos \omega = \langle a, b \rangle$, the cosine of the angle between the axes of two chosen meridians $a, b \in S$.

3. TWO-BRIDGE KNOTS

3.1. Traceless characters are dihedral. Recall Riley’s parametrization [Ril84]: for a two-bridge knot the nonabelian $\mathrm{SL}_2(\mathbb{C})$ representations with meridian eigenvalue s are

$$\rho(a) = \begin{pmatrix} s & 1 \\ 0 & s^{-1} \end{pmatrix}, \quad \rho(b) = \begin{pmatrix} s & 0 \\ -u & s^{-1} \end{pmatrix},$$

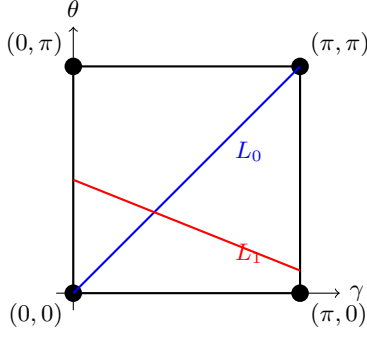


FIGURE 1. The pillowcase $P = (\mathbb{R}/2\pi\mathbb{Z})^2/\iota$, a 2-sphere with four $\mathbb{Z}/2$ orbifold (corner) points, shown on the square $[0, \pi]^2$. For a two-bridge knot the two tangle Lagrangians are straight arcs of rational slope (Section 3.2); being straight they meet transversally and bound no bigon. The earring correspondence bends such an arc into a figure-eight at each corner (Remark 3.2).

cut out by a polynomial $\Phi_{p,q}(s, u)$ of degree $(p-1)/2$ in u ; the traceless locus is $s = i$.

Proof of Theorem 1.1. A binary-dihedral representation sends meridians into the reflection coset $e^{i\phi}j$ of $\text{Pin}(2) = S^1 \cup jS^1$, whose elements are traceless; hence every irreducible dihedral representation is traceless, so there are *at least* $(p-1)/2$ irreducible traceless characters (the number of irreducible metabelian representations of a knot with $|H_1(\Sigma_2)| = p$ odd). On the other hand the traceless Riley polynomial $\Phi_{p,q}(i, u)$ has u -degree at most $(p-1)/2$, so there are *at most* $(p-1)/2$. The bounds coincide, so every irreducible traceless character is one of the dihedral ones. A dihedral character with $\rho(ab)$ of rotation angle $4\pi k/p$ has $\cos \omega = \cos(2\pi k/p)$; distinct $k \in \{1, \dots, (p-1)/2\}$ give distinct characters, and these values depend only on p . Finally $\phi_p(u) = \prod_k (u - u_k)$ with $u_k = 2 \cos(2\pi k/p) - 2 = -4 \sin^2(\pi k/p)$; its constant term is $\prod_k 4 \sin^2(\pi k/p) = p$ and its root-sum is $\sum_k (2 \cos(2\pi k/p) - 2) = -1 - (p-1) = -p$. $\square$

Remark 3.1. Small cases: $\phi_3 = u + 3$, $\phi_5 = u^2 + 5u + 5$, $\phi_7 = u^3 + 7u^2 + 14u + 7$, $\phi_9 = u^4 + 9u^3 + 27u^2 + 30u + 9$. For p prime, ϕ_p is, after $u = 2c - 2$, the minimal polynomial of $2 \cos(2\pi/p)$, irreducible of degree $(p-1)/2$. Theorem 1.1 is elementary and is essentially classical (the traceless/metabelian coincidence for two-bridge knots is implicit in [Ril84]); we include the proof because it fixes the interior generators exactly and q -independently, which is what the pillowcase picture needs. The statement was verified by exact Gaussian-integer computation of the Riley word for all $p \leq 9$, including the two determinant-9 knots $9_1 = \mathfrak{b}(9, 1)$ and the hyperbolic $6_1 = \mathfrak{b}(9, 2)$, which produce the identical root set $\{-4 \sin^2(\pi k/9)\}$.

3.2. The pillowcase consequence and the earring. The character variety of a rational tangle of slope r is a straight arc of slope r in the pillowcase, and the double branched cover of the four-punctured sphere is a torus T^2 on which the tangles lift to straight lines [Lin92, HHK14, CHKK22]. For $\mathfrak{b}(p, q)$ the two lifted lines have classes $(0, 1)$ and (p, q) ; they meet in $|\det \begin{pmatrix} 0 & 1 \\ p & q \end{pmatrix}| = p$ points, and gluing the two solid tori recovers $\Sigma_2(K) = L(p, q)$. Being straight lines, they bound no immersed bigon.

Two things follow, and one warning. First, the naive count on P itself is $(p+1)/2$: the $(p-1)/2$ interior dihedral characters of Theorem 1.1 together with the single abelian character at a corner. The passage to the full $\det = p$ requires the reduction defining $I^\natural$, i.e. the earring. Second, and structurally: the differential and the bounding-cochain deformation are assembled

from holomorphic bigons and discs, of which straight lines have none. Thus on the two-bridge family the figure-eight bubbling that obstructs the general conjecture is inert.

This is consistent with the known outcome: Hedden–Herald–Kirk [HHK14, HHK18] prove that for every two-bridge knot the $\mathbb{Z}/4$ gradings agree and the differentials vanish on both sides, so pillowcase homology is isomorphic to $I^\natural$ with zero bounding cochain, of rank $\det$. We emphasize this is *their* theorem; Theorem 1.1 only clarifies why the interior generators are exactly the dihedral characters and why the obstruction is corner-localized.

Remark 3.2 (Correcting a naive expectation). It is tempting to hope that composing the straight Lagrangians with the Cazassus–Herald–Kirk–Kotelskiy earring correspondence $E \subset P^- \times P$ [CHKK22] preserves straightness. It does not: Herald–Kirk [HK24] prove, at the regular-homotopy level, that E sends an arc-type Lagrangian to a genuine figure-eight curve with one self-crossing per corner, while doubling loop-type curves that avoid the corners. Thus $E \circ L_1$ is immersed, and all of its new geometry is confined to the four corners. For two-bridge knots the net effect is nonetheless trivial (zero bounding cochain suffices), and every known example with nonzero bounding cochain—the $(4, 5)$ -torus knot [CHKK22], the pretzel knot $P(-2, 3, 5)$ [Smi24], and the further pretzel members computed in the companion paper [Wue26]—is non-alternating.

4. THE $(3, n)$ -TORUS KNOTS

4.1. The representation arcs. Write $\pi_1(S^3 \setminus T(3, n)) = \langle x, y \mid x^3 = y^n \rangle$ with $\gcd(3, n) = 1$. The element $z = x^3 = y^n$ is central, so any irreducible $\mathrm{SU}(2)$ representation sends $z \mapsto \pm 1$ and

$$\rho(x) = e^{\alpha \hat{u}}, \quad \rho(y) = e^{\beta \hat{v}}, \quad \alpha = \frac{\pi \ell_1}{3}, \quad \beta = \frac{\pi \ell_2}{n},$$

with $\hat{u}, \hat{v}$ unit imaginary quaternions, $\ell_1 \in \{1, 2\}$, $0 < \ell_2 < n$, and the central-sign match $\ell_1 \equiv \ell_2 \pmod{2}$. The angle $\theta = \angle(\hat{u}, \hat{v}) \in (0, \pi)$ is the only remaining modulus; each such pair (ℓ_1, ℓ_2) is a one-parameter arc of irreducible characters (Klassen [Kla91]), and we call these the *representation arcs*. Call an arc *admissible* if it contains a traceless character; the criterion, derived in the proof of Proposition 4.1 below, is $|\cot(b\beta)| < \sqrt{3}$.

4.2. The traceless locus. The meridian is $\mu = x^a y^b$ with $an + 3b = 1$; it is well defined modulo the center, and tracelessness is insensitive to that ambiguity (multiplying μ by z negates $\rho(\mu)$, preserving trace 0). A direct quaternion computation gives

$$\frac{1}{2} \operatorname{tr} \rho(\mu) = \operatorname{Re} \rho(\mu) = \cos(a\alpha) \cos(b\beta) - \sin(a\alpha) \sin(b\beta) \cos \theta,$$

which is affine in $\cos \theta$. Hence each arc contains exactly one traceless character, at

$$\cos \theta = \cot(a\alpha) \cot(b\beta), \tag{1}$$

provided the right side lies in $(-1, 1)$ (otherwise the arc carries none; the boundary case never occurs, as the proof below shows). We verified in each case that the resulting ρ satisfies $\rho(x)^3 = \rho(y)^n = (-1)^{\ell_1}$ and $\operatorname{tr} \rho(\mu) = 0$ exactly; both central signs occur, and for n odd the traceless characters split evenly between the two sheets (see the proof of Proposition 4.1).

Proposition 4.1 (Count). *The number of irreducible traceless $\mathrm{SU}(2)$ characters of $T(3, n)$ is*

$$N(3, n) = \begin{cases} 2(n-1)/3, & n \equiv 1, \\ (2n-1)/3, & n \equiv 2, \\ (2n+1)/3, & n \equiv 4, \\ 2(n+1)/3, & n \equiv 5, \end{cases} \pmod{6},$$

together with one abelian traceless character; thus $N(3, n) \sim 2n/3$.

Proof. From $an + 3b = 1$ we get $\gcd(a, 3) = \gcd(b, n) = 1$. Since $\ell_1 \in \{1, 2\}$ and $3 \nmid a\ell_1$, the first cotangent in (1) always has $|\cot(a\alpha)| = \cot(\pi/3) = 1/\sqrt{3}$; so an arc is admissible iff $|\cot(b\beta)| < \sqrt{3}$, iff $\pi b\ell_2/n$ lies in $(\pi/6, 5\pi/6)$ modulo π , iff the residue $m = b\ell_2 \bmod n$ lies in the open interval $(n/6, 5n/6)$. Equality never occurs: it would force $6 \mid n$, impossible with $\gcd(3, n) = 1$, so the boundary case $\cos\theta = \pm 1$ does not arise. The parity constraint assigns to each $\ell_2 \in \{1, \dots, n-1\}$ exactly one ℓ_1 ($\ell_1 = 1$ for ℓ_2 odd, $\ell_1 = 2$ for ℓ_2 even), and $\ell_2 \mapsto m = b\ell_2 \bmod n$ is a bijection of $\{1, \dots, n-1\}$; hence

$$N(3, n) = \#\{m \in \mathbb{Z} : n/6 < m < 5n/6\} = \lfloor 5n/6 \rfloor - \lfloor n/6 \rfloor,$$

which evaluates, in the four residues $n \equiv 1, 2, 4, 5 \pmod{6}$, to the stated cases. Distinct arcs give distinct characters, since $\text{tr } \rho(y) = 2\cos(\pi\ell_2/n)$ separates them, and the abelian traceless character (every meridian to a fixed traceless element) is unique up to conjugacy. Finally, for n odd the involution $\ell_2 \mapsto n - \ell_2$ flips the parity of ℓ_2 (hence exchanges the sheets $\ell_1 = 1 \leftrightarrow 2$) and preserves admissibility, since $m \mapsto n - m$ preserves the interval; being fixed-point-free, it splits the $N(3, n)$ traceless characters evenly between the two central-sign sheets, as claimed above. $\square$

The values $N(3, n)$ for $n = 2, 4, 5, 7, 8, 10, 11, 13$ are 1, 3, 4, 4, 5, 7, 8, 8. As a check, the formula was also verified by direct enumeration of the admissible arcs for all $n \leq 25$.

4.3. The dihedral dichotomy.

Proof of Theorem 1.2. In $\text{Pin}(2) = S^1 \cup jS^1$ every element is a rotation (in S^1) or a reflection (in jS^1); reflections are exactly the traceless elements, and an irreducible dihedral representation must send a generator to a reflection. Now $\rho(x)$ has trace $2\cos(\pi\ell_1/3) = \pm 1 \neq 0$, so it is never traceless (as 3 is odd); and $\rho(y)$ is traceless iff $\beta = \pi/2$ iff $\ell_2 = n/2$, possible only for n even. Hence the number of irreducible dihedral traceless characters is 1 if n is even and 0 if n is odd, which equals $(\det T(3, n) - 1)/2$ since $\det T(3, n) = 3$ for n even and 1 for n odd. In particular, for n odd there are no dihedral characters, so every irreducible traceless character is non-dihedral. $\square$

Theorem 1.2 is the exact opposite of Theorem 1.1: two-bridge knots are entirely dihedral, the $(3, n)$ -torus knots with n odd entirely non-dihedral. The non-dihedral characters are precisely those the pillowcase construction represents by non-straight Lagrangians, on which the earring produces genuine figure-eight curves; this is why the family is a natural first test beyond the metabelian world.

Remark 4.2. There is a visible symmetry: $(\ell_1, \ell_2) \mapsto (3 - \ell_1, n - \ell_2)$ preserves θ in (1) (both cotangents change sign), and it respects the parity constraint $\ell_1 \equiv \ell_2 \pmod{2}$ exactly when n is odd; in that case it pairs the two central-sign sheets $\rho(x)^3 = \pm 1$ at equal meridian angle—the even split in the proof of Proposition 4.1.

5. THE $\mathbb{Z}/4$ GRADINGS VIA THE DOUBLE BRANCHED COVER

The double branched cover of $T(3, n)$ is the Brieskorn sphere $\Sigma(2, 3, n)$. For n odd it is an integral homology sphere; its irreducible flat $\text{SO}(3)$ connections have rotation numbers $(m_1, m_2, m_3) = (1, m_2, m_3)$ indexing the three singular fibers, with m_2, m_3 even, $0 < m_2 < 3$, $0 < m_3 < n$. Since 2 is the only even value in $(0, 3)$, necessarily $m_2 = 2$, and m_3 ranges over the even integers in $(0, n)$ meeting the spherical-triangle inequalities on the angles $(\pi/2, 2\pi/3, \pi m_3/n)$. (These m_i index the branched cover and are not the (ℓ_1, ℓ_2) of Section 4.) Their number is $a = -\sigma(T(3, n))/4$, and each lifts to two traceless knot characters, so

$$N(3, n) = 2a \quad (n \text{ odd}), \quad \sigma(T(3, n)) = -2N(3, n), \quad (2)$$

which we verified (independent triangle-inequality count) for all odd $n \leq 43$. Thus the character count of Proposition 4.1 determines the signature and, with the grading formulae below, the graded homology. Throughout this section a relatively $\mathbb{Z}/4$ -graded group is recorded by its rank vector (r_0, r_1, r_2, r_3) , the ranks in gradings 0, 1, 2, 3.

5.1. The grading formula. The $\mathbb{Z}/4$ grading is the spectral flow of the extended Hessian mod 4, relative to the trivial connection Θ (a single generator, in grading $\sigma \bmod 4$); every irreducible contributes four generators, two in grading μ and two in $\mu + 1$ [KM11b, PS17]. The knot grading is

$$\mu(\alpha) = \frac{1}{2} \text{gr}(\alpha) + \frac{1}{4}(1 - \rho_{\text{Ad} \alpha}) \pmod{4},$$

where, for $\Sigma(2, p, q)$ with $p = 3$, $q = n$, the Fintushel–Stern index [FS90, Anv16] and the equivariant ρ -invariant [Anv16] are

$$\text{gr}(\alpha) = \frac{e^2}{pq} + \frac{2}{p} \sum_{k=1}^{p-1} \cot \frac{\pi k}{p} \cot \frac{\pi b_2 k}{p} \sin^2 \frac{\pi k m_2}{p} + \frac{2}{q} \sum_{k=1}^{q-1} \cot \frac{\pi k}{q} \cot \frac{\pi b_3 k}{q} \sin^2 \frac{\pi k m_3}{q}, \quad (3)$$

$$\rho_{\text{Ad} \alpha} = 1 - \frac{2}{pq} - \frac{4}{p} \sum_{k=1}^{p-1} \cot \frac{\pi k}{p} \tan \frac{\pi b_2 k}{p} \cos^2 \frac{\pi k m_2}{p} - \frac{4}{q} \sum_{k=1}^{q-1} \cot \frac{\pi k}{q} \tan \frac{\pi b_3 k}{q} \cos^2 \frac{\pi k m_3}{q}, \quad (4)$$

with $e = pq m_1 + 2q m_2 + 2p m_3$ and b_2, b_3 the even Seifert invariants ($pq b_1 + 2q b_2 + 2p b_3 = 1$); since the cotangents see them only mod p, q , one may take $b_2 \equiv (2q)^{-1}$, $b_3 \equiv (2p)^{-1}$.

Example 5.1. For $\Sigma(2, 3, 7)$ the two connections $(1, 2, 2)$ and $(1, 2, 4)$ give $\text{gr} = 175 \equiv 7$ and $\text{gr} = 251 \equiv 3 \pmod{8}$, $\rho = 3$ in both cases, and $\mu = 87 \equiv 3$ and $\mu = 125 \equiv 1 \pmod{4}$. This agrees with Anvari's Example 6.1 [Anv16] at the level of the unreduced values, confirming our evaluation of (3)–(4).

5.2. The gradings of $T(3, n)$, n odd. Evaluating (3)–(4) over the admissible connections gives the following.

Proposition 5.2. *Let n be odd and coprime to 3, and set $a = -\sigma(T(3, n))/4 = N(3, n)/2$ (an even integer). Under the grading (3), the irreducible connections of $\Sigma(2, 3, n)$ (there are a of them) split evenly: $a/2$ have $\mu = 1$ and $a/2$ have $\mu = 3$ (never 0 or 2). Consequently the reduced singular-instanton chain complex is*

$$IC^\natural(T(3, n)) = (1 + a, a, a, a), \quad \chi = +1,$$

with the trivial generator Θ in grading 0. This is the Poudel–Saveliev chain-level statement [PS17, §7.4] (Anvari [Anv16] computes exactly these generators and gradings; the irreducible part is Daemi–Scaduto's [DS24]). The homology $I^\natural$ then splits by $n \bmod 6$ (here $\sum_i |\Delta|$ is the sum of the absolute values of the coefficients of the Alexander polynomial):

$$\text{rank } I^\natural(T(3, n)) = \sum_i |\Delta_{T(3, n)}| = \begin{cases} 1 + 4a, & n \equiv 1 \pmod{6} \quad (\partial = 0), \\ 4a - 1 = (1 + 4a) - 2, & n \equiv 5 \pmod{6} \quad (\text{rank } \partial = 1). \end{cases}$$

For $n \equiv 1$ the Alexander lower bound [Lim10, KM11a] $\sum_i |\Delta| = 1 + 4a$ meets the chain rank, forcing $\partial = 0$ and $I^\natural = IC^\natural = (1 + a, a, a, a)$ (Anvari [Anv16, Ex. 6.2] proves this family). For $n \equiv 5$ the differential is nonzero; the smallest case is $T(3, 5) = P(-2, 3, 5) = 10_{124}$, where $\text{rank } I^\natural = 7$ (not 9), pinned by the coinciding Alexander and Khovanov bounds, both equal to 7 [Smi24], and where on the pillowcase side the nonzero bounding cochain of Smith [Smi24] enacts the cancellation.

Proof and verification. On $\Sigma(2, 3, n)$ one has $\rho \equiv 3$, so $\mu = (\text{gr} - 1)/2$. A $\mu = 3$ connection contributes two chain generators to each of gradings 3 and 0; a $\mu = 1$ connection two to each of 1 and 2. That a is even follows from $a = (n - 1)/3 = 2m$ for $n = 6m + 1$ and $a = (n + 1)/3 = 2m + 2$ for $n = 6m + 5$; so with $a/2$ of each type the irreducibles give (a, a, a, a) , and Θ adds one to grading 0, giving $IC^\natural = (1 + a, a, a, a)$ of chain rank $1 + 4a$. The even split (hence this chain complex) is proved for $n \equiv 1 \pmod{6}$ by Anvari [Anv16, Ex. 6.2] and verified here by direct evaluation of (3)–(4) for all odd $n \leq 43$. For the homology, $\sum_i |\Delta_{T(3,n)}| = 1 + 4a$ when $n \equiv 1$ and $4a - 1$ when $n \equiv 5$; since the Alexander and Khovanov bounds give $\sum_i |\Delta| \leq \text{rank } I^\natural \leq \text{rank } \widetilde{Kh}$ (rational coefficients) [Lim10, KM11a, KM11b] and, for the $(3, n)$ -torus knots, $\text{rank } \widetilde{Kh} = \sum_i |\Delta|$ — from Turner’s rational Khovanov homology of $(3, q)$ -torus links [Tur08], the unreduced theory being two shifted copies of the reduced one over $\mathbb{Q}$ [Shu14] — the differential vanishes iff $n \equiv 1$; for $n \equiv 5$ it has rank 1. (The case $T(3, 5)$ is Smith’s [Smi24], with both bounds equal to 7.) $\square$

Explicitly $a = (n - 1)/3$ for $n \equiv 1$ and $a = (n + 1)/3$ for $n \equiv 5 \pmod{6}$; either way $\text{rank } I^\natural \sim 2N(3, n)$ grows while $\det = 1$ is fixed, so $I^\natural(T(3, n))$ is far from thin. We note that $\chi(I^\natural(K)) = +1$ for *every* knot [PS17, Thm 8.1]—carried by the single trivial generator—so here χ coincides with $\det = 1$ only by accident; in general $\chi \neq \det$, as the non-thin case 8_{19} below ($\chi = 1$, $\det = 3$) shows.

n	5	7	11	13	17	19	23	25
$n \bmod 6$	5	1	5	1	5	1	5	1
$N(3, n) = 2a$	4	4	8	8	12	12	16	16
$a = -\sigma/4$	2	2	4	4	6	6	8	8
chain rank $1 + 4a$	9	9	17	17	25	25	33	33
rank $I^\natural$	7	9	15	17	23	25	31	33

TABLE 1. For $T(3, n)$, n odd coprime to 3: the traceless character count $N = 2a$, the chain rank $\dim IC^\natural = 1 + 4a$ (Proposition 5.2), and the homology rank $\text{rank } I^\natural = \sum_i |\Delta_{T(3,n)}|$. The two coincide for $n \equiv 1 \pmod{6}$ (differential zero) but differ by 2 for $n \equiv 5 \pmod{6}$ (bold; differential rank 1). $T(3, 5) = P(-2, 3, 5) = 10_{124}$ is rigorous (Alexander and Khovanov bounds both equal 7; [Smi24]); the graded ranks for $n \equiv 1$ are $(1 + a, a, a, a)$.

The Fintushel–Stern grading is special to n odd, where $\Sigma(2, 3, n)$ is a homology sphere; for n even $b_3 \equiv (2p)^{-1} \pmod{n}$ does not exist. A nonzero differential appears both for $n \equiv 5 \pmod{6}$ (above) and for n even. The smallest and cleanest instance—bounding cochain vanishing, differential a single bigon—is $8_{19} = T(3, 4)$, to which we now turn.

6. A NONZERO DIFFERENTIAL: $8_{19} = T(3, 4)$

For n even the double cover $\Sigma(2, 3, n)$ carries nontrivial reducibles ($H_1 \neq 0$), and the pillowcase complex acquires a differential. We treat the smallest case, $8_{19} = T(3, 4)$, in two ways. (Rank vectors (r_0, r_1, r_2, r_3) record ranks in gradings 0, 1, 2, 3, as in Section 5.)

6.1. Structural derivation. By Theorem 1.2 and Proposition 4.1, $T(3, 4)$ has three irreducible traceless characters—two non-dihedral (at $\theta \approx 125.3^\circ, 54.7^\circ$) and one dihedral (at 90°)—together with one abelian character. Via the double cover $\Sigma(2, 3, 4)$, whose first homology is $\mathbb{Z}/3$: the two non-dihedral characters are the two lifts of the single Σ -irreducible; the dihedral character is the nontrivial reducible; the abelian character is Θ . Through the grading framework (an irreducible

contributes four generators, two in grading μ and two in grading $\mu + 1$; the nontrivial reducible two; Θ one, in grading $\sigma \bmod 4 = 2$), with $\mu_{\text{irr}} = 3$ and $\mu_{\text{red}} = 1$, the chain complex assembles to

$$C^\natural = (2, 1, 2, 2), \quad \chi = +1,$$

matching the instanton computation of Poudel–Saveliev [PS17, Ex. 7.9]. A finite search over grading-respecting differentials shows that the *only* differential compatible with homology $(2, 1, 1, 1)$ is a single rank-one map $\partial_3: C_3 \rightarrow C_2$: one bigon from a grading-3 generator (a lift of the irreducible, an interior point) to a grading-2 generator (the reducible/ Θ , a corner point). This derivation is independent of [HHK18], and it predicts an interior-to-corner bigon.

6.2. The explicit bigon. Hedden–Herald–Kirk [HHK18, §11.6] realize exactly this. With L_0 the earring figure-eight and L_1 the perturbed character variety of the torus-knot tangle with the parameters $(p, q, r, s) = (3, 4, 3, -2)$ of [HHK18, §11], the singular unperturbed variety — shaped like the letter φ , a circle with an arc through it, whence the name “ φ -curve” of [HHK18, §11.6] — resolves under perturbation into an arc R_0 and a vertically monotone circle R_1 . The seven generators are r_+ (corner, $\text{gr} = 2$), $x_1^\pm$ (arc), and $x_2^\pm, x_3^\pm$ (circle); the gradings are $\text{gr}(x_1^+) = 0$, $\text{gr}(x_1^-) = 3$, and the circle contributes one generator in each grading. There is exactly one bigon, $x_1^- \rightarrow r_+$, running into the corner along R_0 and the earring, giving $\partial x_1^- = r_+$ —a rank-one map from grading 3 to grading 2. The circle bounds no bigon. The homology is

$$H^\natural = (1, 0, 0, 0)_{\text{arc}} \oplus (1, 1, 1, 1)_{\text{circle}} = (2, 1, 1, 1) = I^\natural(8_{19}),$$

of rank $5 > \det = 3$; the complex is non-thin.

6.3. The corner-bigon thread. The structural derivation and the explicit computation give the same graded complex and the same differential. The bigon is a corner figure-eight bigon—the earring’s self-crossing at a pillowcase corner linking an interior irreducible character to the reducible locus. This is precisely the mechanism absent on two-bridge knots (Section 3.2): there the Lagrangians are straight, there is no corner bigon, and $\partial = 0$; here the single corner bigon is what makes $I^\natural(8_{19})$ non-thin. The geometric computation is due to [HHK18]; our contribution is the independent structural derivation predicting the differential’s rank, source, and target from the character count alone.

7. THE OPEN DIRECTION

Everything above concerns generators and gradings; the analytic heart of the knot Atiyah–Floer conjecture is the bounding cochain. On two-bridge knots it is inert (Section 3.2); off them it is genuinely nonzero, as shown for the $(4, 5)$ -torus knot [CHKK22] and the pretzel knot $P(-2, 3, 5)$ [Smi24]. Notably $P(-2, 3, 5) = T(3, 5) = 10_{124}$ is itself a member of our family—the smallest $n \equiv 5 \pmod{6}$ case—so the nonzero differential of Proposition 5.2 for $n \equiv 5$ is, on the pillowcase side, precisely Smith’s nonzero bounding cochain. A differential thus appears for $n \equiv 5$ (odd) and for all n even; two problems, of increasing difficulty, remain.

First, the instanton side. For every torus knot the irreducible singular instanton homology is $\mathbb{Z}_{(1)}^{\lceil -\sigma/4 \rceil} \oplus \mathbb{Z}_{(3)}^{\lfloor -\sigma/4 \rfloor}$ (subscripts denote the grading, superscripts the rank: the rank vector is $(0, \lceil -\sigma/4 \rceil, 0, \lfloor -\sigma/4 \rfloor)$) with vanishing differential [DS24]. For n even, however, $\Sigma(2, 3, n)$ carries a nontrivial reducible ($\det = 3$, so $H_1 = \mathbb{Z}/3$), and the reduced complex has the shape

$$C^\natural(T(3, n)) = \underbrace{\frac{N-1}{2} \text{ irreducibles}}_{4 \text{ generators each}} \oplus \underbrace{\text{one reducible}}_2 \oplus \Theta, \quad \dim = 2N(3, n) + 1,$$

on which an irreducible-to-reducible differential can appear—exactly the corner figure-eight bigon of 8_{19} (Section 6). Organizing this differential across the even family—in particular, whether it

is always of rank one, as for 8_{19} , giving $I^\natural$ of rank $2N(3, n) - 1$ —is open beyond the single case $n = 4$ [PS17].

Second, and harder, the pillowcase side: to compute the bounding-cochain-deformed pillowcase invariant and match it to $I^\natural$. No bounding cochain has been computed for an infinite family in any setting; before the companion paper the record was the $(4, 5)$ -torus knot (correct rank [CHKK22]) and $P(-2, 3, 5)$ (nonzero only [Smi24]). The companion [Wue26] computes bounding cochains for three members of the pretzel family $P(-2, 3, q)$, within the immersed-curve combinatorial model of [HK24, Smi24]: at $q = 5$ and $q = 7$ the cochain cancels a differential of the undeformed complex, and at $q = 11$ it creates one. The family-wide computation, and the analytic identification, remain open there too. Both problems lie beyond the present, purely representation-theoretic, account.

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